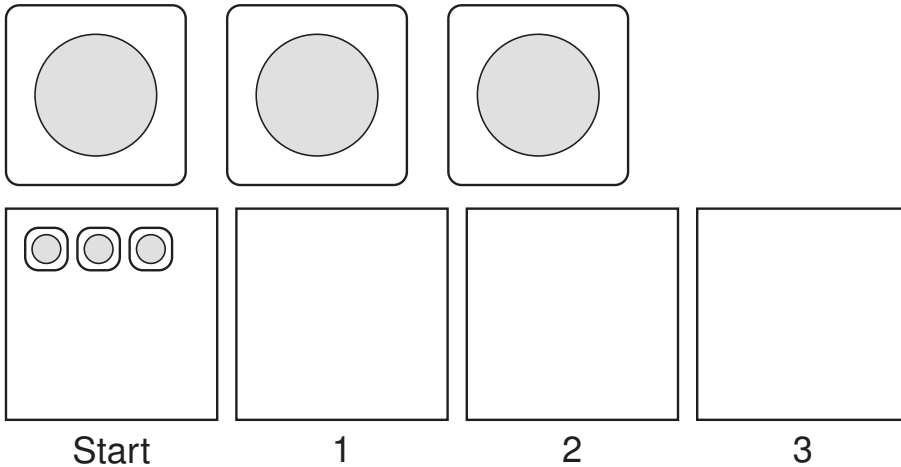
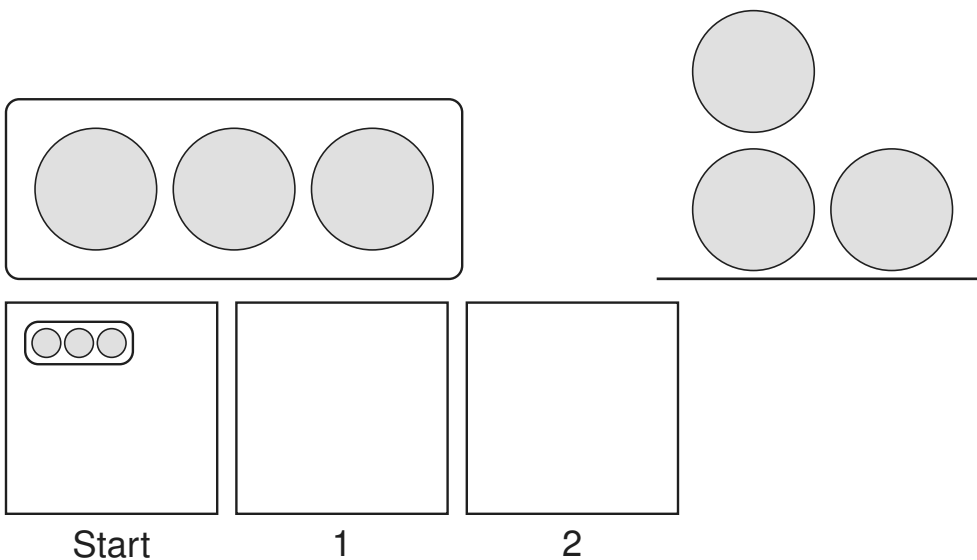


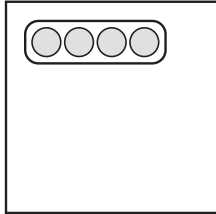
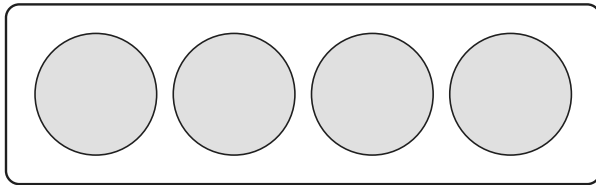
**Problem 1:** Take 3 counters. Put one counter on each circle in the picture: three piles of 1. A move goes like this: take one counter from every pile, and put those counters together as a new pile. Make one move. Draw your piles in box 1: one dot for each counter, and a ring around each pile. Make another move and draw it in box 2. Make one more move and draw it in box 3. Point to the boxes that show the same piles.



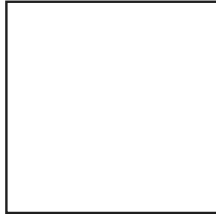
**Problem 2:** Take 3 counters in one pile. Make moves and draw each one, until a box looks the same as the box before it. Then put your counters on the stairs picture: a tower of 2 next to a tower of 1. Do they fit exactly, one counter on each circle and none left over?



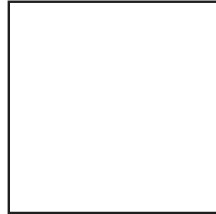
**Problem 3:** Take 4 counters in one pile. Make 5 moves. Draw your piles after each move. Point to the boxes that show the same piles as box 1. Keep making moves. Do the piles ever stop changing?



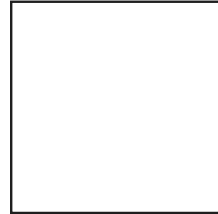
Start



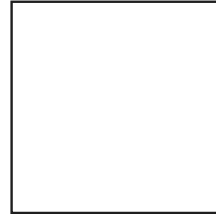
1



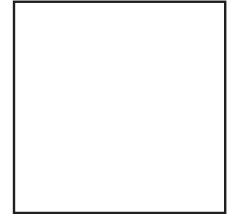
2



3

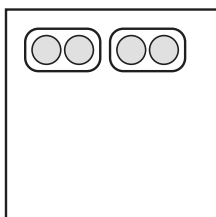
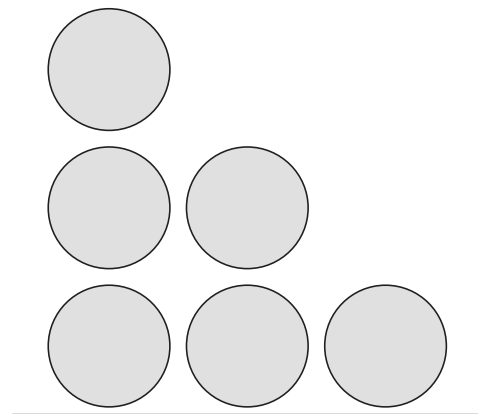
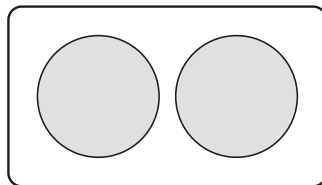
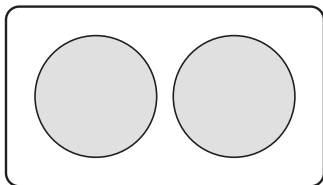


4



5

**Problem 4:** Put 4 counters in two piles of 2. Make 3 moves and draw each one. Which box shows the same piles as the start? Then try to put your 4 counters on the small stairs picture, one on each circle. Try the big stairs picture. Do 4 counters fit exactly on either one?



Start



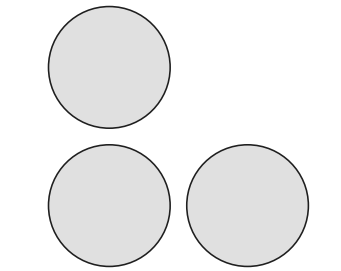
1



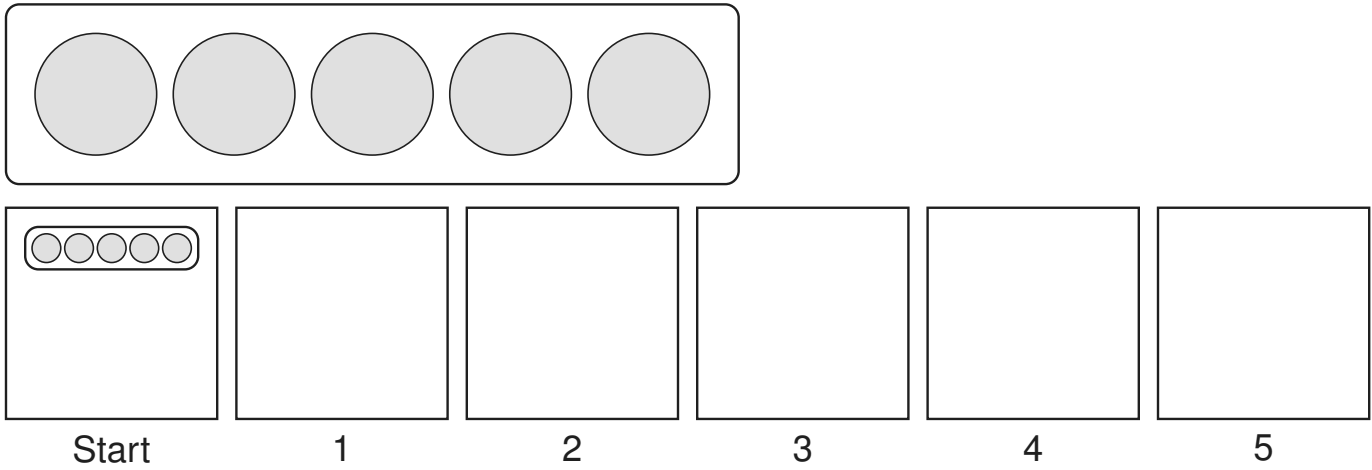
2



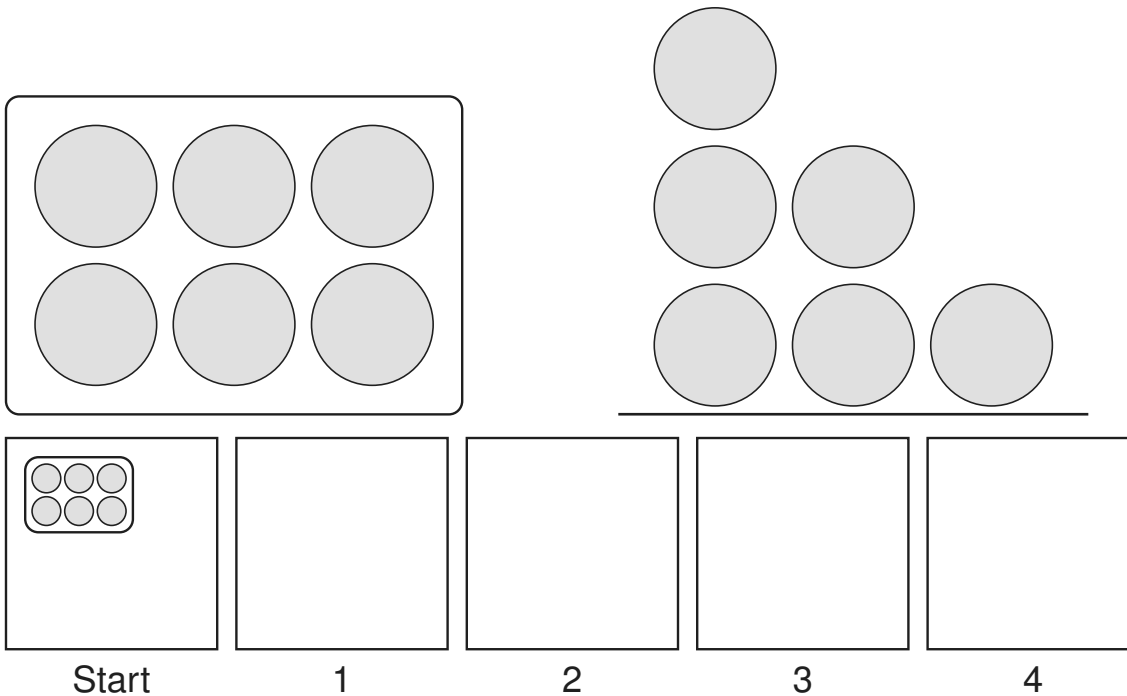
3



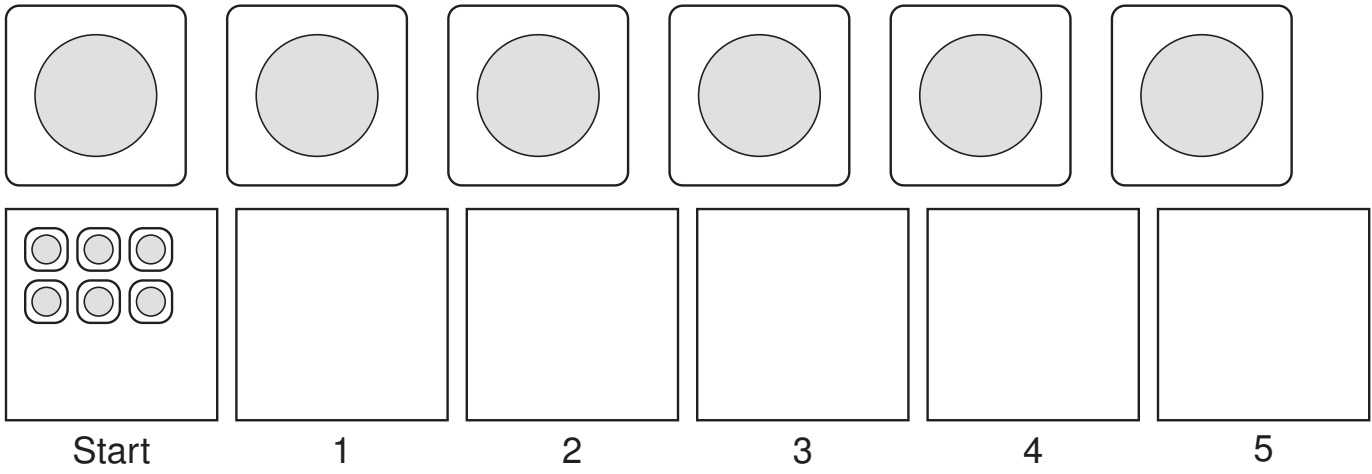
**Problem 5:** Take 5 counters in one pile. Make 5 moves. Draw your piles after each move. Point to the two boxes that show the same piles. Keep making moves. Do the piles ever stop changing?

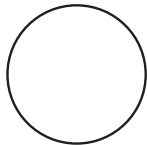


**Problem 6:** Take 6 counters in one pile. Make moves and draw each one, until a box looks the same as the box before it. Then put your counters on the stairs picture with towers of 3, 2 and 1. Do they fit exactly?

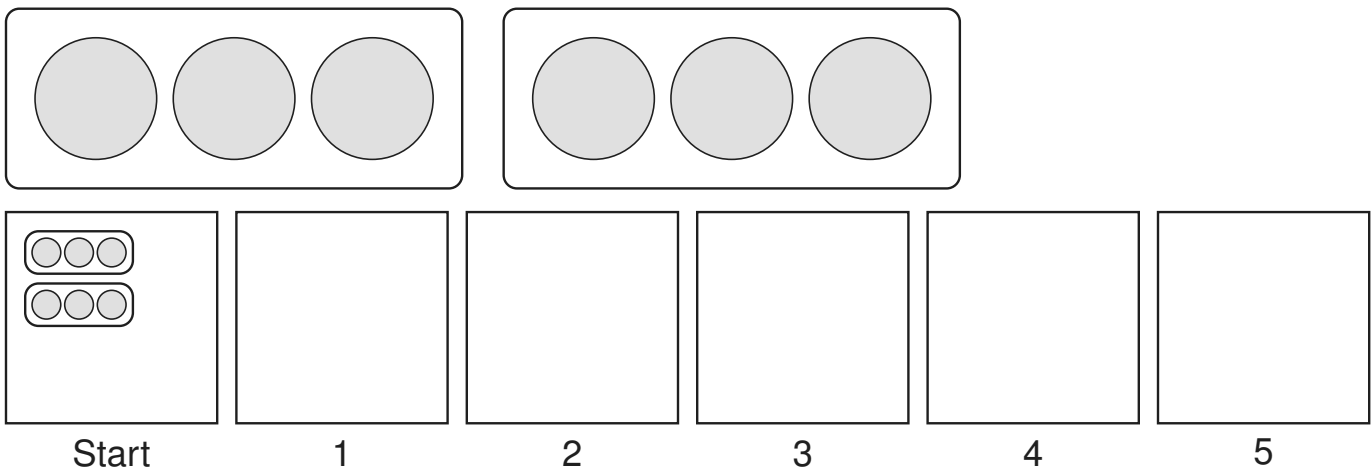


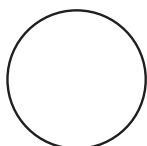
**Problem 7:** Put 6 counters in six piles of 1. Make moves and draw each one, until a box looks the same as the box before it. How many moves did it take to reach the stairs, towers of 3, 2 and 1? Write the number in the circle.



 moves

**Problem 8:** Put 6 counters in two piles of 3. Make moves and draw each one, until a box looks the same as the box before it. How many moves did it take to reach the stairs, towers of 3, 2 and 1? Write the number in the circle.



 moves

**Problem 9:** Put 6 counters in piles of 2, 2, 1 and 1. Make moves and draw each one, until a box looks the same as the box before it. How many moves did it take to reach the stairs? Write the number in the circle. Which of your four games with 6 counters took the most moves?

Start

1

2

3

4

5

6

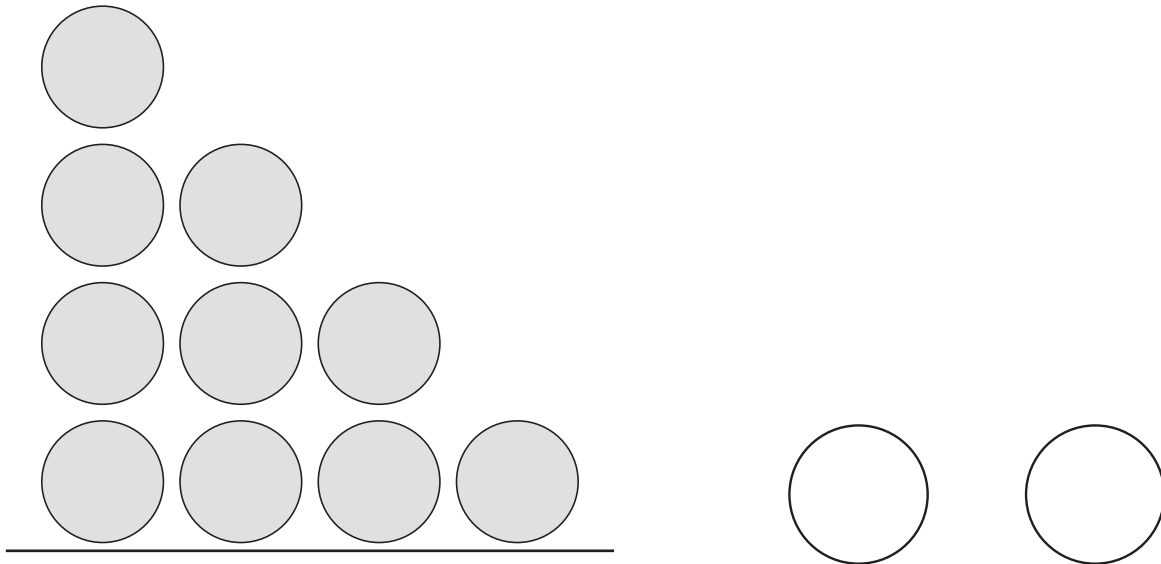
7

moves

**Problem 10:** Look back at your games. For each number of counters, make a check mark in the table: did the piles stop changing, or did they go round and round? Fill in the rows for 7 and for 10 after Problems 12 and 13.

| Counters |  Stopped |  Round and round |
|----------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 3        |                                                                                             |                                                                                                     |
| 4        |                                                                                             |                                                                                                     |
| 5        |                                                                                             |                                                                                                     |
| 6        |                                                                                             |                                                                                                     |
| 7        |                                                                                             |                                                                                                     |
| 10       |                                                                                             |                                                                                                     |

**Problem 11:** Build stairs with 3 counters: a tower of 2 next to a tower of 1. Build stairs with 6 counters: towers of 3, 2 and 1. Now build the next stairs, one step taller, and put them on the big stairs picture. Count the counters. Write the number in the first circle. Then build stairs that are one step taller again. Count those counters. Write the number in the second circle.



**Problem 12:** Take 10 counters in one pile. Make moves and draw each one, until a box looks the same as the box before it. Put your counters on the big stairs picture. Do they fit exactly? Then make a check mark for 10 counters in the table of Problem 10.

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
|       |   |   |   |   |   |
| Start | 1 | 2 | 3 | 4 | 5 |
|       |   |   |   |   |   |
|       |   |   |   |   |   |
| 6     | 7 |   |   |   |   |

**Problem 13:** Take 7 counters in one pile. Make 7 moves. Draw your piles after each move. Point to the boxes that show the same piles as box 3. Do 7 counters stop, or go round and round? Make a check mark for 7 counters in the table of Problem 10.

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
|       |   |   |   |   |   |
| Start | 1 | 2 | 3 | 4 | 5 |

|   |   |
|---|---|
|   |   |
| 6 | 7 |

**Problem 14:** Put 8 counters in four piles of 2. Make 5 moves. Draw your piles after each move. Point to the boxes that show the same piles. How many different pictures does this game go round on?

|  |  |
|--|--|
|  |  |
|  |  |

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
|       |   |   |   |   |   |
| Start | 1 | 2 | 3 | 4 | 5 |